ECO 304: Analytics for Economics

Problem Set 3: Inference for Bivariate Regressions

Purpose: This problem set gives you practice using standard errors, hypothesis tests, p values, and confidence intervals to assess uncertainty in bivariate regression estimates.

Files and submission

Use the supplied folder structure.

- `01_Data/02_Base/` contains the prepared survey and World Bank data used in Problem Sets 1 and 2. It also contains the World Bank codebook. Data cleaning is not part of this problem set.
- `01_Data/01_Raw/` is included to preserve the course folder structure. You do not need to use it.
- `02_Code/` contains `ps3_inference.R`, the R starter script. Update its project path and complete your analysis there. Your code should run from start to finish and create all requested figures and tables.
- Save all tables and figures in `03_Output/`.
- Save your final report in `04_Report/`.

For both sections, use a two-sided test with a 5% significance level. In your report, write the relevant hypothesis, show the requested results in a clearly labeled table, and answer every interpretation question in complete sentences.

Part I. Exercise and sleep in the student survey

Return to `survey_results_clean.csv`. The question is: *Among students in this survey, is exercise time associated with weeknight sleep?*

Use `weeknight_sleep` as the outcome and `exercise_time` as the explanatory variable. Both variables are measured in hours.

1. Write the population model for this question. Let i index students, use `weeknight_sleep` and `exercise_time` in your equation, and include a residual.
2. State the null and two-sided alternative hypotheses for the slope on exercise time.
3. Create a scatterplot with exercise time on the horizontal axis and weeknight sleep on the vertical axis. Add the fitted regression line. Give the figure a descriptive title and clear axis labels.
4. Estimate the bivariate regression in R. In a clearly labeled table, report the number of students used, $\hat{\beta}_1$, its standard error, the t statistic, the two-sided p value, and the 95% confidence interval for $\hat{\beta}_1$.

5. Verify the reported t statistic by calculating

$$t = \frac{\hat{\beta}_1 - 0}{se(\hat{\beta}_1)}.$$

6. At the 5% level, do you reject or fail to reject the null hypothesis? Interpret the p value correctly in this setting.

7. Interpret the 95% confidence interval in words. Does it include zero? What does that tell you about the corresponding two-sided test at the 5% level?

8. Is the estimated relationship necessarily causal? Explain one reason why exercise could affect sleep and one concern that would prevent you from interpreting this regression as causal.

Part II. Education and GDP per capita across countries

Return to `world_bank_2019.csv`. The question is: *Across countries, is secondary-school enrollment associated with GDP per capita?*

Use `gdp_per_capita_2015_usd` as the outcome and `secondary_enrollment_gross_pct` as the explanatory variable. The explanatory variable is measured in percentage points. Consult the codebook if you need the exact variable definitions or units.

1. Write the population model for this question. Let i index countries and include a residual.

2. State the null and two-sided alternative hypotheses for the slope on secondary enrollment.

3. Create a scatterplot with secondary enrollment on the horizontal axis and GDP per capita on the vertical axis. Add the fitted regression line. Give the figure a descriptive title and clear axis labels.

4. Estimate the bivariate regression in R. In a clearly labeled table, report the number of countries used, $\hat{\beta}_1$, its standard error, the t statistic, the two-sided p value, and the 95% confidence interval for $\hat{\beta}_1$.

5. Verify the reported t statistic using the same formula from Part I.

6. At the 5% level, do you reject or fail to reject the null hypothesis? Interpret the p value correctly in this setting.

7. Interpret the 95% confidence interval in words. Explain what one additional percentage point of secondary enrollment means for the unit of $\hat{\beta}_1$.

8. Explain why this country-level regression does not, by itself, establish that increasing secondary enrollment causes GDP per capita to change. Discuss one reverse-causality concern or omitted variable.

Brief conclusion

In one short paragraph, compare what statistical inference adds to the two regression results. In particular, distinguish the estimated association from the uncertainty around that estimate, and explain why statistical significance alone does not establish a causal effect or practical importance.